

Raihane Sidhoum

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EDUCATION

Ecole Polytechnique

Paris, France

Quantitative Economics and Finance

Aug. 2026 – May 2028

- Key subjects: Probability Theory, Time Series, Econometrics, Machine Learning, Game Theory and Applications.

HEC Paris

Paris, France

Grande Ecole Program

Aug. 2024 – May 2027

- Key subjects: Financial Markets, Monetary Economics and Sovereign debt, Financial Economics, Corporate Structuring, Python

Ecole Nationale Polytechnique - ENP

Algiers, Algeria

Grande Ecole Program, Engineering

Aug. 2021 – Aug. 2024

- Key subjects: Machine Learning, Probability, Statistics, Markov Chains, Mathematical and Statistical Modeling, Big Data, Object Oriented Programming

EXPERIENCE

FrontierMath AI Adversarial Tester

Mar. 2026 – May. 2026

Mercor AI – Undisclosed Client

San Francisco, CA (remote contractor)

- Created novel olympiad-style mathematical problems in combinatorics. With difficulty level ranging from 15-50 on the Math Olympiad Hardness Scale (MOHS).
- Stumped State of the Art (SOTA) AI Models using created problems and provided solutions, hints and feedback on the model's errors.

Data Intern

Sep. 2025 – Mar. 2026

SLB (ex-Schlumberger)

Paris, France

- Created two applications to automate processes including a request filling and approval app and a knowledge saving app using PowerApps and PowerAutomate.
- Created automated, regularly running Python scripts to fetch data from company websites, format the outputs and save them in specified locations.
- Utilized statistical learning techniques to forecast revenue for 3 divisions across all operating countries, incorporating country-specific team assumptions into a linear regression model.

Supply Chain Intern

Mar. 2024 – July 2024

SLB (ex-Schlumberger)

Hassi Messaoud/Algiers, Algeria

- Collected data on the loading and unloading of vehicles (+60 operation) in the Domestic Logistics fleet, analyzed sources of delay (dwell time), extracted their root causes, presented corrective actions, and designed a digital solution architecture to ensure real-time monitoring of their effectiveness.
- Determined the optimal daily order for trucks to perform loading and unloading using various metaheuristics, with an objective function based on the total truck time simulated using Monte Carlo methods, projecting a 30% time gain.

EXTRACURRICULAR ACTIVITIES

International Mathematical Olympiad

2018 – Present

- Honourable Mention at the 2019 IMO. Algeria's Contestant 3 at IMO 2018 and Contestant 2 at IMO 2019.
- Coached the Algerian MO team during preparations for various competitions. Specialization : Combinatorics and Euclidean Geometry.
- Coached the Rwandan IMO Team at their April 2026 Camp as well as the advanced class of the African Olympiad Academy (including multiple IMO participants) for the month of April 2026.

Industrial Engineers Club (IEC) –President

2023

- Managed a team of +80 members, a network of +440 Alumni and + \$10000 in resources.
- Organized +10 events with professional, scientific and cultural focus. Key Events : Algeria Data Cup, IEC Business Game, TEDxENP.

LANGUAGES AND TECHNICAL SKILLS

Languages: **English:** Fluent (TOEIC 990) – **French:** Fluent (TCF C2) – **Arabic:** Native

Technical skills: Python, PyTorch, AppsScript, PowerApps, VBA, MS Office

Projects: [Dynamic Volatility Graphing App](#) , [Zero to Hero Implementations](#), [Minimalist Note-Taking App](#)

Interests: [Catan](#) – [Puzzlescript](#) – [Cooking](#) – [Boxing](#)